Ericsson PE Node WHI-PE-14 QoS Congestion Event Root Cause Analysis and Resolution

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_5572

Description

The observed QoS congestion event on Ericsson PE node WHI-PE-14 at Whitefield East POP can be attributed to microbursts, interface tail drops, and QoS queue starvation. This phenomenon is a direct consequence of the PE node's inability to manage the allocated bandwidth efficiently. The underlying protocol mechanics, specifically the Multi-Protocol Label Switching (MPLS) and the Label Distribution Protocol (LDP), play a pivotal role in orchestrating the forwarding behavior of the PE node. The MPLS label stack, which is used to forward packets across the network, can become congested, leading to the tail drop of packets. This, in turn, results in the QoS queue starvation, causing a significant degradation in service quality. Furthermore, the PE node's ability to manage the allocated bandwidth is also affected by the presence of microbursts, which are bursts of high-bandwidth traffic that can overwhelm the PE node's forwarding capabilities. In this context, the observed ALARM_5572 is indicative of a critical congestion event that requires immediate attention to prevent further service degradation.

Pre-requisite Checks

- Perform a site visit to Whitefield East POP to verify the PE node's physical status and ensure that all cables and connections are secure.

- Check the redundant link status to ensure that it is functioning correctly and is not contributing to the congestion event.

- Verify that the PE node's software version is up-to-date and that all necessary software patches have been applied.

Potential Root Causes

Cause	Probability
Microbursts causing interface tail drops	High
QoS queue starvation due to inefficient bandwidth allocation	Medium
Label Distribution Protocol (LDP) issues affecting MPLS label stack	Low

Diagnostic Commands & Expected Outputs

```
Command: show mpls interface, Expected Output: Interface information, including label stack and forwarding behavior, Description: This command displays the MPLS interface information, including the label stack and forwarding behavior. It can help identify if the microbursts are causing interface tail drops.
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```
Command: show qos queue, Expected Output: QoS queue information, including bandwidth allocation and utilization, Description: This command displays the QoS queue information, including the allocated bandwidth and utilization. It can help identify if the QoS queue starvation is due to inefficient bandwidth allocation.

Command: show ldp neighbor, Expected Output: LDP neighbor information, including label distribution and forwarding behavior, Description: This command displays the LDP neighbor information, including label distribution and forwarding behavior. It can help identify if the LDP issues are affecting the MPLS label stack.
```

Resolution Steps

Step 1: Step: Upgrade the PE node's software version to the latest release, Command: software upgrade package apply, Description: This step upgrades the PE node's software version to the latest release, which includes bug fixes and enhancements that can improve the PE node's ability to manage bandwidth and prevent microbursts.

Step 2: Step: Adjust the QoS queue configuration to optimize bandwidth allocation, Command: qos queue adjust , Description: This step adjusts the QoS queue configuration to optimize bandwidth allocation, which can help prevent QoS queue starvation.

Step 3: Step: Verify and adjust the LDP neighbor configuration to ensure proper label distribution, Command: ldp neighbor verify , Description: This step verifies and adjusts the LDP neighbor configuration to ensure proper label distribution, which can help prevent LDP issues affecting the MPLS label stack.

Post-Check Verification

```
Command: show mpls interface, Expected Output: Interface information, including label stack and forwarding behavior, Description: This command displays the MPLS interface information, including the label stack and forwarding behavior. It can help verify if the resolution steps have corrected the microbursts issue.

Command: show qos queue, Expected Output: QoS queue information, including bandwidth allocation and utilization, Description: This command displays the QoS queue information, including the allocated bandwidth and utilization. It can help verify if the resolution steps have corrected the QoS queue starvation issue.

Command: show ldp neighbor, Expected Output: LDP neighbor information, including label distribution and forwarding behavior, Description: This command displays the LDP neighbor information, including label distribution and forwarding behavior. It can help verify if the resolution steps have corrected the LDP issues affecting the MPLS label stack.
```

Roll-back Procedures

- Step: Downgrade the PE node's software version to the previous release, Command: software downgrade package apply, Description: This step downgrades the PE node's software version to the previous release, which can help restore the previous configuration and prevent further service degradation.

- Step: Reset the QoS queue configuration to its original state, Command: qos queue reset , Description: This step resets the QoS queue configuration to its original state, which can help restore the previous configuration and prevent further service degradation.

- Step: Verify and adjust the LDP neighbor configuration to its original state, Command: ldp neighbor verify , Description: This step verifies and adjusts the LDP neighbor configuration to its original state,

which can help restore the previous configuration and prevent further service degradation.